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To cite this article: Merve Özge Algedik Tokyürek, İrem Koç, Ebru Öztürk, Gözde Yazıcı & Hayyam Kıratlı (09 Feb 2026): Changes in inflammatory blood parameters in patients with radiation-related complications due to radiotherapy for uveal melanoma, Clinical and Experimental Optometry, DOI: [10.1080/08164622.2026.2625884](https://doi.org/10.1080/08164622.2026.2625884)

To link to this article: <https://doi.org/10.1080/08164622.2026.2625884>



Published online: 09 Feb 2026.



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RESEARCH



Changes in inflammatory blood parameters in patients with radiation-related complications due to radiotherapy for uveal melanoma

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ABSTRACT

Clinical relevance: Radiation retinopathy is a common cause of visual impairment following ocular radiotherapy, and systemic inflammatory activity may contribute to retinal vascular damage. Understanding inflammatory mechanisms may support earlier identification of patients at risk and improve clinical monitoring strategies.

Background: Radiation retinopathy is a vision-threatening complication of radiotherapy for uveal melanoma and has traditionally been considered a localised microvascular disorder. However, the potential contribution of systemic inflammation to its development remains unclear.

Methods: A retrospective review was undertaken of hospital records of patients diagnosed with radiation retinopathy after stereotactic photon radiotherapy for uveal melanoma. Demographic characteristics, tumour features, and systemic inflammatory blood parameters were evaluated before and after the onset of retinopathy.

Results: A total of 97 patients were included, with 52.6% female participants. Most tumours were classified as T1 or T2. The median time to retinopathy diagnosis was 24 months. All systemic inflammatory blood parameters showed significant increases after the development of radiation retinopathy (all $p < 0.05$).

Conclusion: It is concluded that radiation retinopathy is associated with systemic inflammatory activation, suggesting that inflammatory processes may extend beyond localised retinal injury, and therefore may serve as accessible biomarkers for disease monitoring.

ARTICLE HISTORY

Received 22 October 2025

Revised 26 January 2026

Accepted 27 January 2026

KEYWORDS

Ocular radiotherapy;
radiation retinopathy;
radiation toxicity; retinal
vascular injury; systemic
inflammation

Introduction

Radiation therapy is utilised in numerous orbital, peri-orbital, and intraocular tumours in ophthalmology, with uveal melanoma being its principal indication. The primary radiotherapy modalities utilised for uveal melanoma include plaque brachytherapy, stereotactic photon-based radiotherapy or radiosurgery, and particle-based radiotherapy.¹ Each modality, regardless of technical differences, carries inherent risks of radiation toxicity in addition to its therapeutic efficacy. These toxic effects arise from both direct DNA damage and indirect injury mediated by radiation-induced free radicals.² These effects predominantly involve vascular endothelial and pericyte cells. Posterior segment toxicities, mainly radiation retinopathy, are primarily attributed to the consequent vascular damage arising from injury to these cells.

Beyond direct cellular injury, ionising radiation triggers secondary biological responses, including cytokine release due to tumour necrosis, oxidative stress, and activation of inflammatory cascades. Consequently, a proinflammatory milieu develops in the ocular environment following radiotherapy for uveal melanoma, particularly during the early stages.³ Over time, these radiation-related alterations may result in chronic inflammation, ischaemia, and, in severe cases, inadvertent enucleation.

Radiotherapy is also known to induce systemic inflammatory responses, contributing to an increased inflammatory burden that extends beyond the irradiated field.⁴ Parallel to this, haemogram-derived inflammatory biomarkers – including the neutrophil-to-lymphocyte ratio (NLR), monocyte-to-lymphocyte ratio (MLR), platelet-to-lymphocyte ratio (PLR), lymphocyte-to-monocyte ratio (LMR), systemic inflammatory

response index (SIRI), and systemic immune-inflammation index (SII) – have been widely studied in various inflammatory, metabolic, infectious, and malignant conditions.^{5–7}

Elevated neutrophil-to-lymphocyte ratio, monocyte-to-lymphocyte ratio and platelet-to-lymphocyte ratio have been reported in disorders such as inflammatory bowel disease, diabetes mellitus, thyroiditis, gastro-intestinal diseases, cardiovascular disease, and COVID-19 infection, illustrating their sensitivity to systemic inflammatory shifts.^{8–13} Likewise, increased systemic immune system index values have been demonstrated in metabolic and musculoskeletal inflammatory disorders. These markers have also been used for prognosis across many cancer types.^{14–16}

Taken together, these findings support the rationale for investigating whether radiation-induced ocular toxicity is accompanied by measurable changes in systemic inflammatory indices. While these biomarkers have been extensively evaluated in a wide range of systemic diseases, it remains unclear whether radiation toxicity in uveal melanoma – being confined to a localised anatomical region – provokes detectable systemic alterations in haemogram-derived indices. Therefore, this study was conducted to observe the potential systemic effects of radiation-induced pro-inflammatory changes by comparing systemic inflammation parameters measured before and after the development of radiation toxicity.

Methods

This retrospective study included patients who underwent stereotactic radiotherapy from 2020 to 2023 due to uveal melanoma, who subsequently developed radiation toxicity predominantly in the posterior segment during ophthalmological follow-ups. Stereotactic radiotherapy has been applied as previously published.^{17,18}

Evaluated data including: patient age, sex, notable systemic conditions, tumour dimensions, fundoscopic tumour characteristics, date of stereotactic radiotherapy for uveal melanoma, date of detection of radiation toxicity, presentation type of radiation toxicity, individual complete blood count values, and inflammatory indices including neutrophil-lymphocyte ratio (NLR), monocyte-lymphocyte ratio (MLR), platelet-lymphocyte ratio (PLR), systemic inflammatory response index (SIRI), and systemic immune-inflammation index (SII) both pre- and post-detection of radiation toxicity.

Time intervals were recorded between detection of radiation toxicity and blood count assessments conducted pre-toxicity onset and post-onset, as well as total radiotherapy dose (Gy), radiation dose to the optic disc, and radiation dose to the lens (Gy). The tumours were classified according to the 8th Edition of American Joint Committee on Cancer Classification. The presentation pattern of radiation toxicity was classified into four groups based on the predominant clinical condition at presentation: vitreous haemorrhage, radiation retinopathy, radiation papillopathy or neovascular glaucoma.

Radiation retinopathy was further staged as proposed by Finger and Kurli previously).¹⁹ Patients with known rheumatological diseases, blood dyscrasias, or diagnoses that could lead to changes in systemic inflammatory parameters were excluded from the study. The study adhered to the tenets of the Declaration of Helsinki and approved by the local institutional ethics committee (Hacettepe University, project ID: SBA24/031)

Statistical analyses included descriptive statistics indicating mean followed by standard deviation for data with normal distribution and percentages for frequencies. Overall descriptive statistics were reported. Assumptions for the inferential statistics tests among variables were initially verified. Kolmogorov-Smirnov test was employed to test for normal distribution assumption, and variables with test results yielding $p > 0.05$ were considered normally distributed.

For variables that did not meet the assumptions of parametric tests, assumption checks were conducted for non-parametric alternatives. Accordingly, dependent (paired) sample t-tests were used to compare the first and second blood laboratory values of patients in the study group for variable pairs exhibiting normal distribution, while non-parametric dependent (paired) Wilcoxon signed-rank tests were employed for those not showing normal distribution.

Multiple regression analysis was employed in this study to examine variables that could potentially influence changes in the primary outcome indices (neutrophil-lymphocyte ratio (NLR), monocyte-lymphocyte ratio (MLR), platelet-lymphocyte ratio (PLR), systemic inflammatory response index (SIRI) and systemic immune-inflammation index (SII). The variables tested for their effects in the regression analysis

included: age, gender, tumour dimensions (largest basal diameter $\times$ height in millimetres), tumour location (posterior uveal, ciliary body, ciliochoroidal), tumour pigmentation (melanotic/amelanotic), stage of radiation retinopathy according to Finger and Kurli (Stage 1, 2, 3, 4 or 5), presentation of radiation toxicity (radiation papillopathy, neovascular glaucoma, radiation retinopathy, or vitreous haemorrhage), T stage according to the AJCC Classification system, lens radiation dose, optic disc radiation dose, time interval from initial blood draw to radiation retinopathy (months), time interval from diagnosis of radiation retinopathy to second blood draw (months), time interval from initial radiation therapy to blood draw (months), history of hypertension, and history of diabetes.

All analyses conducted in the study were examined within a 95% confidence interval. Analyses were performed on IBM SPSS Statistics for Windows, version 23 (IBM Corp., Armonk, N.Y., USA).

Results

Ninety-seven patients were included in the study. Fifty-one (52.58%) of the patient group were female, and forty-six (47.42%) were male. The mean age was 58.6 years, with no statistically significant difference detected between genders in terms of age (57.1 ± 1.8 years for females and 59.1 ± 1.7 years for males, respectively; $p = 0.4867$). The most common location for the tumour was posterior uvea including juxtapapillary location in 91 patients (93.8%). The tumours most commonly comprised T1 ($n = 33$, 34.0%) and T2 tumours ($n = 39$, 40.2%). The mean largest tumour diameter was 9.1 ± 3.1 and the mean tumour height was 6.0 ± 2.3 mm. The other details regarding patient and tumour characteristics are provided in Table 1.

The predominant presentation of radiation toxicity at onset included radiation retinopathy in 36 (37.1%) patients, vitreous haemorrhage in 41 (42.3%) patients, neovascular glaucoma in 12 (12.4%) patients, and radiation papillopathy in 8 (8.2%) patients. According to the classification of radiation retinopathy stages in the literature, 3 patients (3.09%) were classified as stage 1, 8 (8.25%) as stage 2, 19 (19.59%) as stage 3, and 36 (37.11%) as stage 4. Thirty one patients (31.96%) could not be staged due to the location of the tumour involving the macula.¹² The median dose of total radiation applied was 60 Gy in 3 fractions and the mean lens dose and optic disk doses were 22.0 ± 17.6 Gy and 30.5 ± 18.4 , respectively. The median time from radiotherapy to diagnosis of posterior segment radiation toxicity was 24 months (range: 6 - 120 months).

Table 1. Patient- and tumour- related details and their frequencies. HT: systemic hypertension, DM: type 2 diabetes mellitus, AJCC: American Joint Committee on Cancer.

Variable	Number of patients (<i>n</i>)	Frequency (%)
Gender		
Female	51	51.6%
Male	46	47.4%
Age		
30–39 years	8	8.3
40–49 years	24	14.4
50–59 years	27	27.8
60–69 years	27	27.8
70–79 years	20	20.6
80–89 years	1	1.0
HT	24	24.7
DM	23	23.7
HT + DM	10	10.3
Tumour location		
Posterior uvea other than juxtapapillary location	78	80.4
Juxtapapillary location	12	13.4
Ciliochoroidal	4	4.1
Ciliary body	2	2.1
Tumour appearance		
Melanotic	85	87.6
Amelanotic	23	12.3
T stages according to AJCC Classification		
T1	33	34.0
T2	39	40.2
T3	18	18.6
T4	1	1.0

The mean durations between initial blood counts of patients and radiotherapy, between their initial blood counts and the development of radiation toxicity, between the diagnosis of radiation toxicity and their second blood counts, and between their first and second blood counts were 15.1 ± 21.8 months, 16.0 ± 13.2 months, 9.3 ± 11.1 months, and 26.2 ± 17.4 months, respectively.

When individual parameters of complete blood count, and derived indices including neutrophil-lymphocyte ratio (NLR), monocyte-lymphocyte ratio (MLR), platelet-lymphocyte ratio (PLR), systemic inflammatory response index (SIRI) and systemic immune-inflammation index (SII) were evaluated: haemoglobin, haematocrit (HTC), mean corpuscular haemoglobin concentration (MCHC), leucocyte count, lymphocyte percentage, neutrophil percentage, platelet count, mean platelet volume (MPV), and platelet-lymphocyte ratio (PLR) were assessed using paired t-test. Among these, a significant decrease in haematocrit (HTC) ($p = 0.03$), significant increases in lymphocyte percentage and neutrophil percentage, mean platelet volume (MPV) and platelet-lymphocyte ratio (PLR) ($p < 0.001$, $p < 0.001$, $p = 0.01$, $p = 0.02$, respectively), were observed.

The rest of the tested complete blood count parameters which included: erythrocyte count, mean corpuscular volume (MCV), mean corpuscular haemoglobin (MCH), lymphocyte count, monocyte count, neutrophil count, eosinophil count, basophil count, monocyte percentage, eosinophil percentage, basophil percentage, plateletcrit (PCT), platelet distribution width (PDW), red cell distribution width (RDW), neutrophil-lymphocyte ratio (NLR), monocyte-lymphocyte ratio (MLR), lymphocyte-monocyte ratio (LMR), platelet-lymphocyte ratio (PLR), systemic inflammatory response index (SIRI) and systemic immune-inflammation index (SII) were compared between two blood counts using Wilcoxon signed-rank test.

Among the tested variables, erythrocyte count, lymphocyte count, monocyte count, neutrophil count, basophil percentage, plateletcrit (PCT), neutrophil-lymphocyte ratio (NLR), monocyte-lymphocyte ratio (MLR), lymphocyte-monocyte ratio LMR, platelet-lymphocyte ratio (PLR), systemic inflammatory response index (SIRI) and systemic immune-inflammation index (SII) showed statistically significant changes as shown in Table 2. The changes observed in the main outcome measures of the study, namely neutrophil-lymphocyte ratio (NLR), monocyte-lymphocyte ratio (MLR), platelet-lymphocyte ratio (PLR), systemic inflammatory response index (SIRI), and systemic immune-inflammation index (SII) variables, are depicted in the graph in Figure 1.

In regression analysis, a significant model ($p < .05$) was achieved only for the systemic immune-inflammation index (SII) among the six outcome indices where the effects of variables were tested. In this study, multiple regression analysis was employed to examine relationships between the dependent variable (change in systemic immune-inflammation index), and various independent variables which are listed in the Methods section. The regression model was found to be statistically significant overall, with the included variables explaining 44.3% of the variance in the dependent variable ($R^2 = 0.443$). The overall significance of the model was confirmed through ANOVA testing ($F(22, 49) = 1.769$, $p = 0.049$).

The results concerning the significance levels of the independent variables in the model are as follows; patients with a history of diabetes exhibited significantly higher difference in systemic immune-inflammation index (SII) values compared to those without such history ($B = 159.480$, $p = 0.019$). Similarly,

Table 2. Variables showing statistically significant differences between the two blood counts in the Wilcoxon signed-rank test. Positive z-values indicate an increase in the second measurement, while negative values indicate a decrease. NLR: neutrophil-lymphocyte ratio, MLR: monocyte-lymphocyte ratio, LMR: lymphocyte-monocyte ratio, PLR: platelet-lymphocyte ratio, SIRI: systemic inflammatory response index, SII: systemic immune-inflammation index.

Variable	Z value	p value	Median value in the first blood count	Median value in the second blood count
Erythrocyte count	-2.06	0.04	5.0	4.9
Lymphocyte count	-2.51	0.01	2.2	2.1
Monocyte count	2.32	0.02	0.5	0.52
Neutrophil count	2.74	0.01	4.1	4.3
Basophil percentage	2.47	0.01	0.02	0.03
Plateletcrit	2.02	0.04	0.22	0.22
NLR	4.23	0.00	1.80	2.06
MLR	4.29	0.00	0.22	0.25
LMR	-4.58	0.00	4.57	3.95
PLR	4.4	0.00	114.0	122.0
SIRI	4.32	0.00	0.91	1.11
SII	3.66	0.00	446.7	519.4

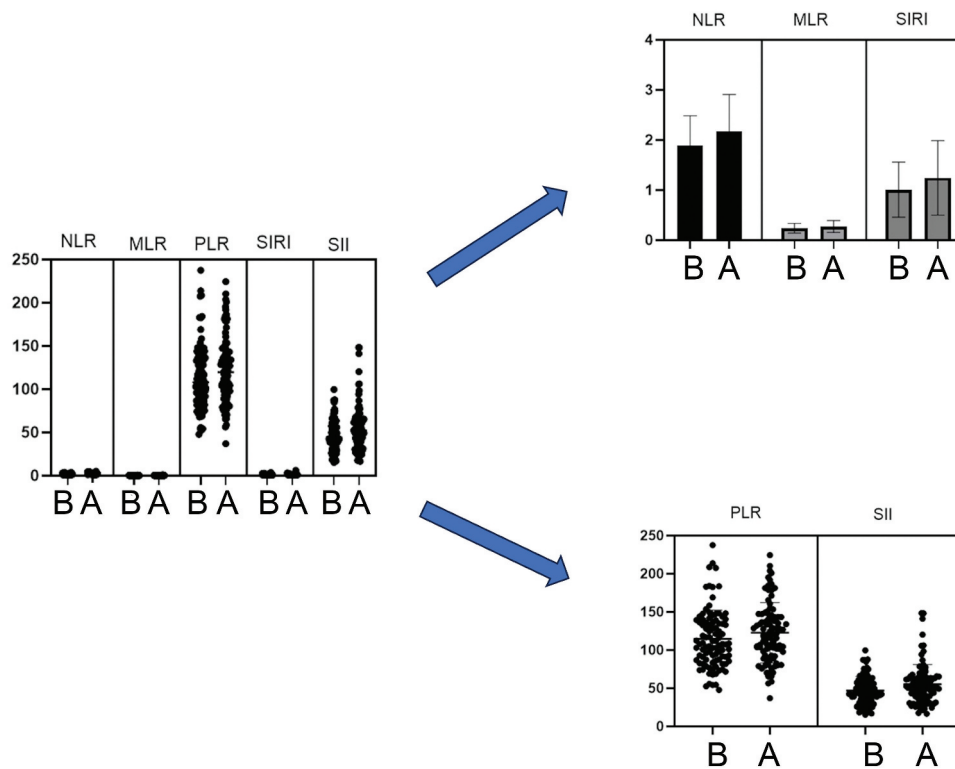


Figure 1. Scatterplot and bar graphs depicting differences of absolute values in the indices before and after detection of radiation retinopathy. B: before detection of radiation toxicity, A: after detection of radiation toxicity, NLR: neutrophil-lymphocyte ratio, MLR: monocyte-lymphocyte ratio, LMR: lymphocyte-monocyte ratio, PLR: platelet-lymphocyte ratio, SIRI: systemic inflammatory response index, SII: systemic immune-inflammation index. Due to the numerical higher values corresponding to PLR and SII indices compared to NLR, MLR, and SIRI, separate figures on the right hand side were created to visualise this relationship.

patients in the Stage 3 category of radiation retinopathy showed significantly higher difference in systemic immune-inflammation index (SII) values compared to Stage 1 disease ($B = 374.671$, $p = 0.032$). None of the other tested independent variables demonstrated a significant effect ($p > 0.05$ for all other mentioned variables).

Discussion

From the perspective of cancer biology, inflammation gains significance due to its role in creating a pro-carcinogenic environment, providing a milieu that supports cancer cell survival and proliferation, and resulting in a microenvironment conducive to immunotherapy.²⁰ The inflammatory microenvironment of tumours is characterised by the presence of host leucocytes within both the stroma and at the tumour sites themselves. These tumour-infiltrating leucocytes can contribute to cancer progression by promoting tumour growth and dissemination, facilitating malignancy-associated immunosuppression, and modulating responses to hormones and chemotherapeutic agents.²¹ Although inflammatory mediators and immune cells play crucial roles in the early phases of tumour development, they are also instrumental in driving malignant cell migration, invasion, and metastasis.²²

Throughout malignant transformation, chemokine receptors and their ligands influence cellular motility, invasiveness, and survival. Upon undergoing oncogenic changes, many cells begin to express chemokine receptors, allowing them to exploit chemokine gradients for migration to, and survival in, distant tissues.²³ For instance, the expression of CCR9 has been linked to melanoma metastasis to the small intestine. Malignant melanoma cells express a broad array of these receptors, which may in part explain the aggressive metastatic potential characteristic of melanoma.²⁴

Studies in the literature have demonstrated that certain inflammation-related markers identified in the biology of uveal melanoma hold prognostic significance. As a recent finding, multi-omics studies have shown that in uveal melanoma, inflammatory age-clock – related exhibit high scores in gene set expression analysis, which are associated with unfavourable prognosis.²⁵ In another study, cytokine and protein expression in the aqueous humour of 90 uveal melanoma cases subjected to enucleation was examined and consequently it was identified that tumours associated with different survival levels expressed three distinct cytokine clusters.²⁶ The highest cytokine expression was associated with the worst prognosis, in addition to advanced patient age, larger tumour sizes, ciliary body involvement, monosomy 3, 8q gain, and the tumours higher stage in AJCC classification.²⁶

Monosomy 3 and Class 2 tumours in the gene expression profile have been associated with a worse prognosis in uveal melanoma of the inflammatory sub-type.²⁷ The present study investigates changes in systemic inflammatory parameters in peripheral blood before and after the onset of radiation-related posterior segment complications, focusing on dynamic changes rather than initial values potentially influenced by tumour characteristics.

Apart from the prognostic impact of inflammation in uveal melanoma biology, there is a pro-inflammatory effect of the applied treatment, namely radiotherapy. Radiation-damaged tissues commonly exhibit hallmark features of inflammation.²⁸ Although the initial physico-chemical events and free radical bursts triggered by radiation occur within microseconds, the subsequent inflammatory response persists, driven by recurring waves of reactive oxygen species (ROS), cytokines, chemokines, and growth factors, accompanied by inflammatory cell infiltration.^{28,29} Beyond its local effects, radiation-induced inflammation significantly influences adaptive, antigen-specific immune responses, shedding light on the complex dynamics of tumour – host interactions and the potential emergence of radiation-induced autoimmune phenomena during therapy.²⁸

The study primarily demonstrates that all six inflammatory indices – neutrophil-to-lymphocyte ratio (NLR), monocyte-to-lymphocyte ratio (MLR), lymphocyte-to-monocyte ratio (LMR), platelet-to-lymphocyte ratio (PLR), systemic inflammatory response index (SIRI), and systemic immune-inflammation index (SII) – increase when compared before and after the development of radiation toxicity findings. This finding suggests that radiation toxicity develops in conjunction with or results in a systemic condition. However, due to the lack of data on the regular monitoring of these indices following radiotherapy for uveal melanoma, it is not possible to definitively conclude whether the observed increase in these indices is solely attributable to the development of radiation toxicity, represents part of an overall response to radiotherapy, or is simply related to the presence of melanoma itself.

The mechanisms underlying the development of radiation retinopathy remain incompletely understood. Current evidence suggests that radiation primarily targets the endothelial cells of the retinal capillaries, where focal endothelial cell loss initiates a cascade of microvascular changes, including aneurysm formation, vascular occlusions, structural abnormalities, and damage characterised by areas of non-perfusion and vascular leakage.³⁰ Although the role of retinal inflammation following radiotherapy, and its contribution to microvascular injury in radiation retinopathy, has yet to be thoroughly investigated, clinical observations indicate that anti-inflammatory agents such as corticosteroids may alleviate symptoms of the condition.³¹

In an animal study by Lebon et al., progressive retinal inflammation was observed post-radiotherapy, suggesting a potential pathogenic role in the development of irreversible retinal and vascular damage.³² It is currently established that the effects of radiotherapy on ocular structures are intricately linked with inflammation.^{32,33} Although radiotherapy applied to the eye is considered a localised treatment, it can have systemic effects. The mechanisms potentially responsible for this local-to-systemic transition can be suggested as: the role of circulating cytokines, increased interaction with systemic circulation due to radiotherapy-induced vascular damage, acute phase response, and radiotherapy-related immune system modulation.^{34,35}

One of the key findings of the present study is that the increase observed in systemic immune-inflammation index (SII) is statistically higher in multivariate analysis in Stage 3 compared to Stage 1 according to Finger and Kurli staging, and in patients with diabetes compared to those without.¹⁹ Accordingly, given that macular oedema or retinal neovascularization is a primary finding in Stage 3, the development of macular oedema or neovascularization may also indicate a more severe systemic inflammation. While the presence of diabetes was identified as a risk factor for increased systemic

inflammation in the present study, the diabetic control at the precise time of toxicity development in patients was not known, precluding correlation with the severity of diabetes and the degree of increase in systemic immune-inflammation index (SII). Inflammation has been widely considered to be interlinked with Diabetes Mellitus, and the complex relationship between the two has been further confirmed, with increased SII correlating with an elevated risk of Diabetes Mellitus, as well as diabetic retinopathy.^{36,37}

The systemic inflammatory response index (SIRI) and the systemic immune-inflammation index (SII) are utilised as they consider more than two blood components, thus providing a broader assessment. These components are platelets, neutrophils, and lymphocytes for systemic immune-inflammation index (SII), and monocytes, neutrophils, and lymphocytes for systemic inflammatory response index (SIRI). In the present study, in multiple regression analysis, a significant model could only be established for changes in systemic immune-inflammation index (SII) among the indices tested in these analyses. The primary reason for this limitation is the insufficient sample size to establish statistically significant models for each index individually, together with the need to evaluate multiple clinical factors that may independently influence these indices. Therefore, the discussion primarily focuses on the significant effects observed through the systemic immune-inflammation index (SII).

Although the prognostic significance of the systemic inflammatory response index (SIRI) and systemic immune-inflammation index (SII) indices has been extensively studied across various cancer types, fewer studies have evaluated their association with secondary effects induced by radiation therapy. Emerging evidence suggests that systemic inflammation markers may play a role in predicting radiation-related toxicities. For instance, in patients with locally advanced nasopharyngeal carcinoma, elevated pre-treatment systemic immune-inflammation index (SII) levels were found to be significantly associated with an increased risk of developing severe radiation-induced oral mucositis, highlighting the potential of systemic immune-inflammation index (SII) as a predictive biomarker for acute inflammatory toxicities during radiotherapy.³⁸

Similarly, in small cell lung cancer patients, systemic inflammatory response index (SIRI) and systemic immune-inflammation index (SII) have been investigated as predictors of radiation oesophagitis, with higher index values correlating with a greater likelihood of experiencing this adverse event.³⁹ These findings suggest that systemic inflammatory response index (SIRI) and systemic immune-inflammation index (SII), beyond their prognostic implications, may also serve as accessible, cost-effective tools for identifying patients at elevated risk of developing radiation-induced inflammatory complications. Nonetheless, further research is needed to validate these observations in specific contexts such as radiation retinopathy, where data remain scarce.

These observations support the present findings, which demonstrate a significant increase in systemic inflammatory indices such as systemic immune-inflammation index (SII) and systemic inflammatory response index (SIRI) following the diagnosis of radiation retinopathy. Although radiation retinopathy is a relatively rare and delayed complication of radiotherapy, the elevated inflammatory markers observed in this cohort suggest a potential role of systemic inflammation in its pathogenesis. Taken together with prior studies on radiation-induced mucositis and oesophagitis, these results further strengthen the hypothesis that systemic inflammatory indices may serve as early, non-invasive biomarkers for radiation-induced toxicities across different organ systems. However, given the limited data specifically addressing ocular complications, prospective studies are warranted to validate the utility of SII and SIRI in predicting or monitoring radiation retinopathy.

Systemic inflammatory blood parameters were assessed in two groups: before and after the onset of radiation retinopathy in patients who had undergone radiotherapy. The observed increase in these indices following the diagnosis of radiation retinopathy may reflect a progressively evolving inflammatory milieu induced by radiation exposure. Regardless of the underlying cause, the analysis demonstrated a consistent elevation in inflammatory markers after the development of radiation retinopathy, suggesting a potential role of inflammation in its pathogenesis.

The retrospective nature of the present study introduces several limitations. The timing of complete blood count measurements could not be fully standardised. To minimise temporal variability, for each patient the blood count obtained closest to the onset of radiation retinopathy was deliberately selected; nevertheless, some degree of timing-related bias cannot be entirely excluded.

Several potential confounders – such as infections, systemic inflammation, corticosteroid use, and comorbidities – could not be systematically controlled and may have influenced inflammatory indices independently of radiation retinopathy.

Detailed radiation dose parameters (including retinal dose, apex dose, and fractionation schedule) were not consistently available, which prevented a reliable assessment of potential dose – response relationships regarding systemic inflammatory changes.

Additional markers of systemic inflammation, such as C-reactive protein (CRP) and erythrocyte sedimentation rate (ESR), were not evaluated due to inherent constraints of the study design, which also limits causal inference and reduces the generalisability of the findings.

Although the increases in inflammatory indices reached statistical significance, their clinical relevance remains uncertain. The present study cannot determine whether these biomarker changes have any impact on visual outcomes, disease progression, or the need for therapeutic intervention. Therefore, the findings should be interpreted as hypothesis-generating, and further longitudinal studies are required to establish whether systemic inflammatory responses meaningfully influence the clinical course of radiation retinopathy. Future prospective studies with standardised blood sampling, comprehensive dosimetric data, and broader inflammatory profiling are warranted to clarify these associations and strengthen the interpretability of these results.

In conclusion, although statistically significant alterations in peripheral blood markers were observed following radiation, the clinical relevance of these changes remains uncertain. Nevertheless, characterising these systemic, non-ocular effects of radiation may help refine future risk stratification efforts and guide more individualised post-treatment monitoring strategies.

Disclosure statement

No potential conflict of interest was reported by the author(s).

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